Physics Isomorphic Questions

Vectors and One-Dimensional Kinematics

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A survey robot starts at the origin of an xy-coordinate system and detects a beacon at position $\vec{r} = -8\hat{i} + 15\hat{j}$, where distances are measured in meters. Which of the following statements is correct?

A. The distance between the beacon and the origin is 23 m.
This comes from adding component magnitudes, $8 + 15$, instead of finding the vector magnitude. Distance from the origin must be computed with the Pythagorean relation, not by direct addition of components.

B. The distance between the beacon and the origin is 17 m.
The distance from the origin is the magnitude of the position vector: $|\vec{r}| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17\,\text{m}$. The sign of the x-component does not make the distance negative; only the squared components matter in the magnitude formula.

C. The distance between the beacon and the origin is 15 m.
This uses only the y-component and ignores the horizontal displacement. That error treats one leg of the right triangle as if it were the full distance.

D. The distance between the beacon and the origin is 8 m.
This uses only the magnitude of the x-component and ignores the y-component. The full distance must include both perpendicular components.

Answer: *B*

Q2. The displacement of a particle moving along the x-axis is given by $\vec{s}(t) = \left[(0.80\,\text{m/s}^3)t^3 - (1.90\,\text{m/s}^2)t^2\right]\hat{i}$. Which of the following statements is correct?

A. This motion has constant acceleration.
This is the common mistake of assuming any polynomial position function produces constant acceleration. Differentiating twice gives an acceleration that depends on t, so it is not constant.

B. The magnitude of the acceleration is $1.90\,\text{m/s}^2$.
This incorrectly reads a coefficient from the displacement function as the acceleration itself. Acceleration must be obtained from the second derivative of displacement.

C. The magnitude of the acceleration at $t = 0$ is $3.8\,\text{m/s}^2$.
Differentiate twice: $s(t) = 0.80t^3 - 1.90t^2$, so $v(t) = 2.40t^2 - 3.80t$ and $a(t) = 4.80t - 3.80$. At $t = 0$, the acceleration is $-3.80\,\text{m/s}^2$, so its magnitude is $3.8\,\text{m/s}^2$.

D. The magnitude of the acceleration at $t = 0$ is $0\,\text{m/s}^2$.
This usually comes from thinking the cubic term makes the acceleration vanish at $t = 0$. However, the t^2 term contributes a nonzero constant to the second derivative.

Answer: *C*

Q3. A probe moves for 4.0 s from position $\vec{r}_A = -7\hat{i} + 1\hat{j}$ m to position $\vec{r}_B = 5\hat{i} - 8\hat{j}$ m. Which of the following statements is correct?

A. The moving distance is 15 m.
15 m is the magnitude of the displacement, since $|\Delta\vec{r}| = \sqrt{12^2 + (-9)^2} = 15$ m. The actual distance traveled depends on the path taken, which is not given.

B. The speed at $t = 0$ must be 3.75 m/s.
This divides displacement magnitude by elapsed time and then mistakes that result for the initial speed. Average rate based on endpoints does not determine the instantaneous speed at the start.

C. The acceleration at $t = 0$ must be 0.94 m/s^2.
This assigns a unique acceleration from only two positions and a time interval. Many different motions with different accelerations could connect the same endpoints in 4.0 s.

D. None of the above is correct.
The displacement is $\Delta\vec{r} = (5-(-7))\hat{i} + (-8-1)\hat{j} = 12\hat{i} - 9\hat{j}$, whose magnitude is $|\Delta\vec{r}| = \sqrt{12^2 + (-9)^2} = 15$ m. But that value is not necessarily the distance traveled, and neither the initial speed nor the acceleration at $t = 0$ can be determined uniquely from only the initial position, final position, and elapsed time.

Answer: *D*